HW1

1. To verify P defined here is a probability, we need to make sure it satisfy the axioms of probability.:

① Axiom 1 : $0 \le P(A)$ for any subset $A \subseteq \Omega$;

② $P(\Omega)=1$;

③ For any infinitely long sequence of disjointed subset $\{A_i\}_{i=1}^{\infty}$, i.e., $A_i \cap A_j = \emptyset$ if $i \neq j$, we have

$$P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$$

For ① $P(A)$ is the ratio of the number of element in A to the number of element in Ω. The smallest value of $P(A)$ is when A is an empty set, so that $P(A) = \frac{0}{n} = 0$. Therefore, Axiom 1 is satisfied.

For ② The number of elements in Ω is n, so $P(\Omega) = \frac{n}{n} = 1$, axiom 2 is satisfied.

For ③ We have to check $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$, $\bigcup_{i=1}^{\infty} A_i = A_1 \cup A_2 \cup A_3 \cup \cdots$
For disjoint sets $A_1, A_2, \cdots$, since they are disjoint, the number of elements in their union is the sum of the number of elements in each set.: $\#(\bigcup_{i=1}^{\infty} A_i) = \#(A_1) + \#(A_2) + \cdots$
So we have $P(\bigcup_{i=1}^{\infty} A_i) = (\#(A_1) + \#(A_2) + \cdots)/n$
Based on the definition of $P(A)$, we now have

$$P(\bigcup_{i=1}^{\infty} A_i) = P(A_1) + P(A_2) + P(A_3) + \cdots$$
$$= \sum_{i=1}^{\infty} P(A_i), \text{ Axiom 3 is satisfied.}$$

① Therefore, $P(A) = \#A/n$ satisfies all three axioms of probability.

HW1

2.

iv) Since $A \subseteq \Omega$, and the result of iii),

We prove $P(A) \leq P(\Omega) = 1$

v) Since $A \cup B = A \cup (B - AB)$, and $A(B-AB) = \emptyset$,

$AB \subset B$, also the results from iii) and ii),

we have

$$P(A \cup B) = P(A) + P(B - AB)$$

$$= P(A) + P(B) - P(AB)$$

$$= P(A) + P(B) - P(A \cap B)$$

vi) ~~[illegible]~~

We prove vi) by induction.

Base case: For $n = 2$, from already proved other properties,

we know that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\leq P(A) + P(B) \quad (\text{because } P(A \cap B) \geq 0)$$

Induction Hypothesis: Suppose this is true for n events,

$P(A_1 \cup \cdots \cup A_n) \leq \sum_{i=1}^{n} P(A_i)$

Induction steps: We want to show it for $n+1$.

$$P(A_1 \cup \cdots A_n \cup A_{n+1}) = P((A_1 \cup \cdots \cup A_n) \cup A_{n+1})$$

$$= P(A_1 \cup \cdots \cup A_n) + P(A_{n+1})$$

$$\leq \sum_{i=1}^{n} P(A_i) + P(A_{n+1})$$

$$= \sum_{i=1}^{n+1} P(A_i)$$

HW 1.

3.

(a) $P(A) = P(w \in \Omega \mid X(w) = 1.5)$

Since $X(w) = w+1$,

We have $1.5 = w+1 \Rightarrow w = 0.5$

Based on the set up of the question, we know $X(w)$ is a continuous random variable in the interval $[0, 1]$, the probability of a specific point is zero. in a continous distribution. Therefore $P(A) = 0$.

(b) The event is not an impossible ~~event~~ event. Instead, it is a zero probability event.

(c) $P(B) = P(w \in \Omega \mid X(w) = 0.5)$

Since $X(w) = w+1 \Rightarrow 0.5 = w+1 \Rightarrow w = -0.5$. This is contraditory to that ~~Given that~~ $w \in \Omega = [0, 1]$, Therefore $B = \emptyset$. So $P(B) = 0$.

(D) Yes, event B has been explained in (C) that it is actually an impossible event.

(e) In order to calculate the CDF of X, we have to find $P(X \leq x)$ for all x in the range of X.

Since $X(w) = w+1$, $\Rightarrow P(X \leq x) = P(w+1 \leq x) \Rightarrow P(w \leq x-1)$.

The range of $X(w)$ is in the interval $[0, 1]$, so we have

$P(X \leq x) = P(0 \leq x-1)$ for x in $[0, 1]$

Specifically,

① if $x < 0$, $P(0 \leq x-1) = P(0 \leq$ a negative number$) = 0$. This is because $X(w)$ cannot be smaller than 0 in the given interval.

② if $0 \leq x < 1$, $P(0 \leq x-1) = P(0 \leq$ a non-negative number$) = 1$. This is because $X(w)$ is always bigger than or equal to 0 in the given interval.

③

③ if $x \geq 1$, $P(0 \leq x-1) = P(0 \leq$ a positive number$) = 1$, since $X(w)$ is always greater than zero in the given interval.

Therefore, the CDF of X is:

$$F(x) = \begin{cases} 0, & x < 0 \\ 1, & 0 \leq x < 1 \\ 1 & x \geq 1 \end{cases}$$

4. The CDF of a random variable X, denote as $F_X(t)$, is defined as:

$$F_X(t) = P(X \leq t)$$

We consider two values t_1 & t_2, where $t_1 \leq t_2$.

① When $t_1 < t_2$,

By the definition of CDF, we have

$$F_X(t_1) = P(X \leq t_1)$$

$$F_X(t_2) = P(X \leq t_2)$$

Given that $t_1 < t_2$, the event $\{X \leq t_1\}$ is a subset of the event $\{X \leq t_2\}$, in other words, $\{X \leq t_1\} \subseteq \{X \leq t_2\}$ event

By the properties of probability measures, if A is a subset of event B, then we have $P(A) \leq P(B)$, then we have

$$P(X \leq t_1) \leq P(X \leq t_2)$$

② When $t_1 = t_2$.

$F_X(t_1) = P(X \leq t_1)$, $F_X(t_2) = P(X \leq t_2)$

Because $t_1 = t_2$, $\{X \leq t_1\}$ and $\{X \leq t_2\}$ actually represent the same event. Therefore, $P(X \leq t_1) = P(X \leq t_2)$, $F_X(t_1) = F_X(t_2)$

In conclusion, we showed that $F_X(t_1) \leq F_X(t_2)$ for all $t_1 \leq t_2$.

We proved that the CDF $F_X(t)$ of X is a nondecreasing function.

(4)